

Prioteasa Cristi Andrei

Data and Computer Science M.Sc. Student - University of Heidelberg

Focus area: Computer Vision and Machine Learning

About Me

I am a self-motivated, purpose-oriented person who prefers getting a high-level general overview of the problem I am dealing with, before I dig into computation and implementation.

Technical Skills	Python, NumPy, Pytorch, Matplotlib, Pandas, Linux (Debian), SQL , LangChain, Elasticsearch
Other Skills	Jira, SCRUM, AGILE, Data Analysis, Linear Algebra, SLAM
Languages	Romanian (Native), English (C2 certified), German (A2 certified), French (A2 *not certified)

Contact

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Education

University of Heidelberg

Heidelberg, Deutschland (2023 - ongoing)

Sponsored through a DAAD Master Stipendium (2023-2025)

- Department: Fakultät für Mathematik und Informatik, Data and Computer Science, M.Sc.
- Coursework: Inverse Problems, CV (3D reconstruction), Time Series Analysis, NLP using Transformers, Generative NN for the sciences, ML Essentials, Computational Statistics, Software Economics

3D UPB Summer School

Bucharest, Romania (Summer 2022)

- Attended workshops on Reinforcement Learning and Traditional Computer Vision Techniques for 12 weeks, for a total of over 360 hours
- As final project, I implemented a Deep Q Learning Agent that plays the game of snake with fixed obstacles using pyTorch and a Planar Segmentation from depth images algorithm using OpenCV in C++
- Coursework: Reinforcement Learning, Tabular and Deep Q Learning, Image Filtering , Depth recovery, Planar Segmentation, Edge Detection, Normal map construction based on the ZED stereo camera, Pattern Matching

Technische Universität Darmstadt

Darmstadt, Deutschland (2021–2022)

Exchange student, Erasmus+ program

- Coursework: Machine Learning for Natural Language Processing, Introduction to Computer Vision, Distributed Systems and Algorithms, Mobile Networks, German Language

Politehnica University of Bucharest Bucharest, Romania (2019–Present)

- Department: Computer Science. Specialization: Machine Learning and Artificial Intelligence
- Coursework: Linear Algebra, Statistics and Probability, Calculus II and III, Data Structures and Algorithms, Numerical Methods, Physics, Functional and OO Programming, Communication Protocols, Digital and Analogues Electronics, Database Administration, x86 architecture, Operating Systems, Computing Systems Architecture, Artificial Intelligence, Introduction to Cryptology, Machine Learning, Computer Vision, Signal Processing

“Fratii Buzesti” National College Craiova, Romania (2015–2019)

- Mathematics and Informatics, English bilingual profile
- Average: 9.46 (out of 10)
- Member of the Dolj County Center of Excellence (https://www.cjexdolj.ro/), studying Mathematics
- Participated in the National Olympiad of Mathematics (2 years) and various Mathematics competitions winning several prizes

Work Experience

Institute for Mathematics, Image & Pattern Analysis Group, University of Heidelberg

Hilfswissenschaftler Heidelberg, Deutschland (February 2024 - May 2024)

Description:

Contributed to Go-Droid-SLAM, currently still unpublished. This is a deep-learning-based dense visual SLAM framework that achieves real-time global optimization of poses and 3D reconstruction. This is achieved by adding a loop closure mechanism using the Co-Visibility matrix to the DROID-SLAM framework. On top of this, we add the full optimization kernel from DROID-Calib, which supports arbitrary camera models and optimizes the camera intrinsics on top of the map and pose graph.

- Responsible for result evaluation, Weights & Biases integration, literature research, running experiments

NetRom Software

Summer Internship: Full stack web developer intern Craiova, Romania (Summer 2021)

Description:

During 2 months, I helped develop a web based consumer application using React and .NET 5, whose purpose was to offer the members of a company a way to enroll, save, sort and attend multiple events organized by the company such as: team buildings, seminars, trainings and TED talks and to offer the managers and organizers of the events a way to grade, check attendance, give feedback and issue diplomas to each participant

- Dealt with problems such as: authentication, security, consistent design across devices and browsers, performance and refactoring code and functionality based on the changing need of the product owner
- Got an insight in the SCRUM methodology, working in sprints and having regular meetings with the product owner and the team technical lead. Developed new teamwork abilities and gained practical experience with Jira.

Projects

Artistic colorization with Adaptive Instance Normalization (SS 2024)

- Explored a joint method for the colorization of gray-scale images and color style transfer. We propose an architecture consisting of two parts: a style en- coder based on a VGG19 and a UNet, tasked with colorization and enhanced by Adaptive Instance Normalization layers. Both the weights of the AdaIN layers and of the style encoder, on which they rely upon, are trained implicitly. Moreover, we explore the influence of the chosen colorization loss (MSE vs LPIPS), the chosen colorspace (RGB vs LAB) and conduct an ablation study on different configurations of the network. Finally we show examples of colorization and color transfer from some paintings.

AI Explainability for Human Action Recognition based on skeletal joints motion (Bachelor Thesis) (Winter 2022)

- Under the guidance of the Artificial Intelligence and Machine Learning Department of the Faculty of Automatic Control and Computer Science of University Politehnica of Bucharest I analyzed and implemented techniques for visual explanation in the context of Human Action Recognition based on skeletal joint motion. I explored different state of the art approaches used in Computer Vision such as Class Activation Mapping (CAM), Gradient-weighted Class Activation Mapping (Grad-CAM) and Guided Backpropagation and applied them in the problem of Human Action Recognition. I explored the importance of interpretability, the relationship between faithfulness and interpretability and how to use the intuition gained from Grad-CAM visualizations to detect and correct biases in the dataset.

Deep Reinforcement learning methods (SS 24)

- Implemented reinforcement learning algorithms (DQN, tabulan Q learning, A2C), trained using pyTorch, for the game of Bomberman and Snake
- Learned how to construct, train and analyze a pyTorch model based on reinforcement learning as well as hyperparameter fine-tuning for this specific problem. Experimented with *stable baseline* and *gymnasium*.

RAG system on Pubmed data (SS 24)

- Developed a Question Answering system based on the Retrieval-Augmented Generation (RAG) architecture, utilizing ElasticSearch for information retrieval and LangChain for the RAG framework. Articles from PubMed were scraped, cleaned, and used as the document corpus. The study involved a detailed analysis of various RAG components and their influence on system performance. Additionally, we implemented and evaluated different performance metrics and explored failure cases to assess the system's robustness.

Image based steganography and cryptography (Winter 2022)

- Implemented and analyzed the results of Bloisi, Domenico & Iocchi, Luca. (2007) paper "Image based steganography and cryptography." using Python and NumPy.

Please consult my other projects on my github page.

Interests

I enjoy getting educated on Cognitive Science, philosophy and psychiatry. I play several instruments and love debating.

Voluntary work

Board of European Student of Technology (BEST) Bucharest, Romania (2019–2021)

BEST is an international organization present all across Europe whose goal is to bring together students in an effort to overcome their limits and achieve sky-high goals.

- Observer since the first year of my bachelor's degree
- Member of the Design Team, attending meetings, creating models, banners and taking decisions about the image of various events such as hackathons

Epilepsy Challenge Marathon Craiova, Romania (2018)

- Proud participant and supporter of Epilepsy Challenge, a 5 km cross organized with the purpose to bring awareness and reduce social stigma targeted towards persons who suffer from epilepsy and raise funds for new medical equipment.

Heidelberg, Deutschland - Prioteasa Cristi Andrei, 30 August 2024